

시초 : image classification 문제 해결
image data

Convolutional Networks

Gwangsu Kim

JBNU Dep. of Stat.

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Special topics for Machine Learning

Outline

Intro.

Motivations and Pooling

Variants of the Basic Convolution Function

Structured Outputs and Data Types

Advanced Topics

Intro. I

- Convolution operation

$$s(t) = \int w(t-a)x(a)da, \quad s(t) = (x * w)(t)$$

- Discrete

$$s(t) = \sum_{a=-\infty}^{\infty} w(t-a)x(a)$$

- 2-dimensional conv.

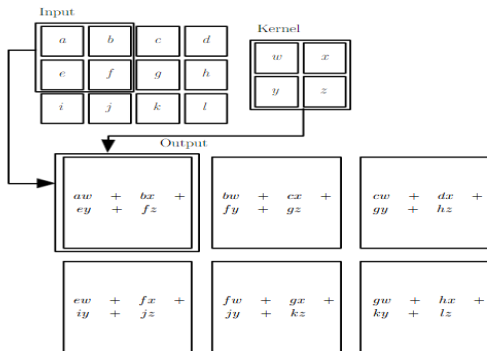
$$S(i,j) = (I * K)(i,j) = \sum_m \sum_n K(i-m, j-n)I(m,n)$$

Intro. II

- DNN uses the cross-correlation formula:

$$S(i,j) = (K * I)(i,j) = \sum_m \sum_n I(i+m, j+n)K(m,n)$$

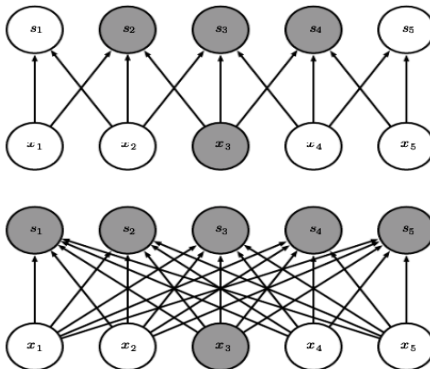
- **Doubly block circulant matrix.**



Motivations and Pooling I

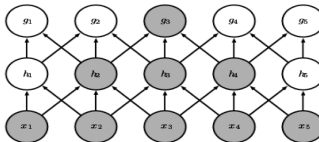
1. Main motivations

- Sparse interactions, parameter sharing, and equivariant representations.

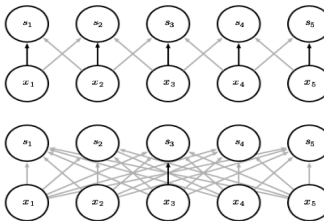


Motivations and Pooling II

- Receptive field.

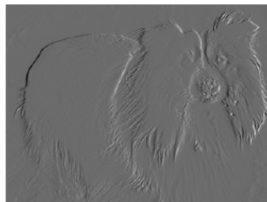


- Parameter sharing: not depending on each pixel.



Motivations and Pooling III

- ▶ Power of edge detection.

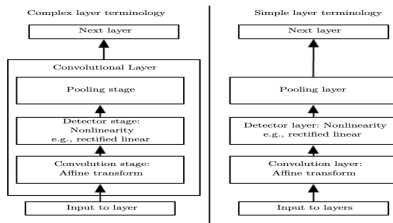


Deep Learning Textbook-Photo credit: PaulaGoodfellow.

- ▶ "Equivariant" of conv.: if f is equivariant to g , $f(g(x)) = g(f(x))$.
 1. Shifting.
 2. Scaling & rotation (non-equiv.).

Motivations and Pooling IV

- ▶ After the conv. operation:
- ▶ **detector stage:** Activation step.
- ▶ **pooling stage:** Using pooling function.

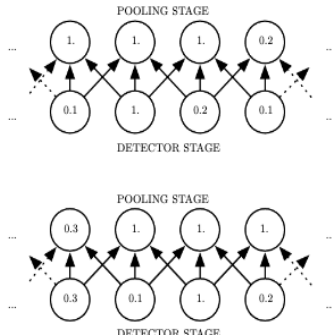


- ▶ The 'convolution layer' terminology can be used for only one layer or combined three layers.

Motivations and Pooling V

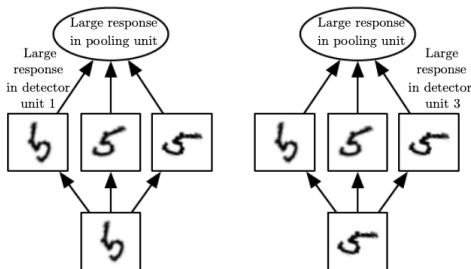
2. Pooling

- ▶ Max pooling: Using the maximums in the partitioned regions.
- ▶ Average pooling: Using the averages in the partitioned regions.



Motivations and Pooling VI

- ▶ In all cases, pooling helps to make the representation approximately invariant to small translations of the input (edge and location finding).
- ▶ The use of pooling can be viewed as adding an infinitely strong prior that the function the layer learns must be invariant to small translations.

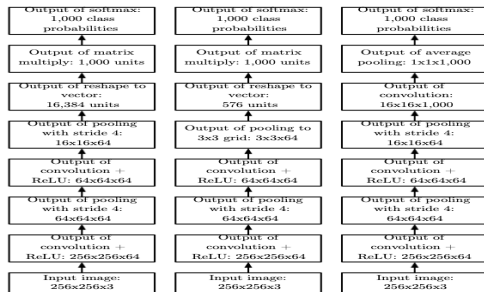


Example of learned invariances.

Motivations and Pooling VII

► Note.

1. Reducing the number of parameters.
2. Fitting to fixed size by the pooling.
3. Dynamic pooling such as using the clustering alg.



Motivations and Pooling VIII

3. Convolution and Pooling as an Infinitely Strong Prior

- ▶ An infinitely strong prior places zero probability on some parameters and says that these parameter values are completely forbidden, regardless of how much support the data give to those values.
- ▶ One key insight understanding by priors is that convolution and pooling can cause underfitting.
- ▶ Another key insight from this view is that we should only compare convolutional models to other convolutional models in benchmarks of statistical learning performance.

Variants of the Basic Convolution Function I

1. Tensor

- ▶ Considering the 4-D kernel tensor, $\mathbf{K}$ with $K_{i,j,k,l}$.

1. Input $\mathbf{V}$ with $V_{i,j,k}$ and output $\mathbf{Z}$ with $Z_{i,j,k}$ where i is the index for channels.

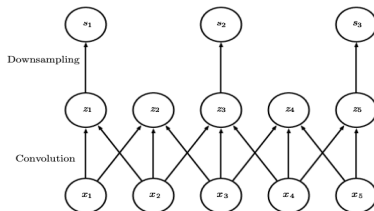
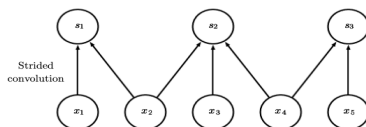
$$Z_{i,j,k} = \sum_{l,m,n} V_{l,j+m-1,k+n-1} K_{i,l,m,n}$$

- 3D image kernel for the multiple channel outputs.

2. Stride k .

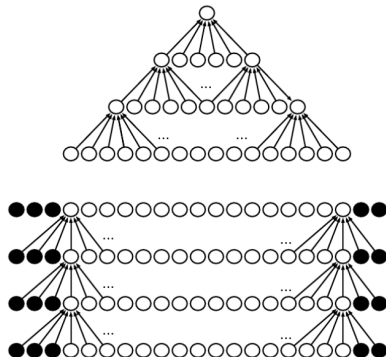
$$Z_{i,j,k} = \sum_{l,m,n} V_{l,(j-1)s+m,(k-1)s+n} K_{i,l,m,n}$$

Variants of the Basic Convolution Function II



Variants of the Basic Convolution Function III

- Padding: adding the lines in the outer of width and height.



Effect of padding.

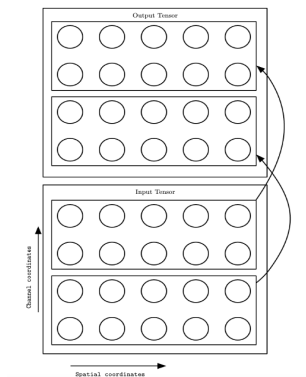
Variants of the Basic Convolution Function IV

- ▶ Locally connected layers (unshared convolution).

$$Z_{i,j,k} = \sum_{l,m,n} V_{l,j+m-1,k+n-1} W_{i,j,k,l,m,n}$$

- ▶ Variant using the partial channel correspondence.

Variants of the Basic Convolution Function V

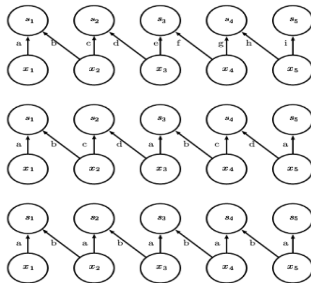


Channel separation.

Variants of the Basic Convolution Function VI

- Tiled convolution.

$$Z_{i,j,k} = \sum_{l,m,n} V_{l,j+m-1,k+n-1} K_{i,l,m,n,j \% t+1,k \% t+1}$$



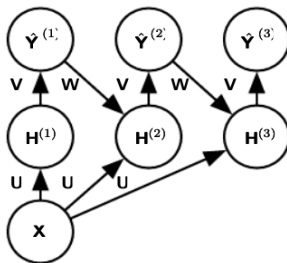
Locally / Tiled / Conventional.

Variants of the Basic Convolution Function VII

- ▶ **Back prop.** (read the textbook).

Structured Outputs and Data Types I

1. Structured Output



Recurrent convolutional network.

Structured Outputs and Data Types II

2. Data Types

	Single channel	Multichannel
1-D	Audio waveform: The axis we convolve over corresponds to time. We discretize time and measure the amplitude of the waveform once per time step.	Skeleton animation data: Animations of 3-D computer-rendered characters are generated by altering the pose of a “skeleton” overtime. At each point in time, the pose of the character is described by a specification of the angles of each of the joints in the character’s skeleton. Each channel in the data we feed to the convolutional model represents the angle about one axis of one joint.

Structured Outputs and Data Types III

	Single channel	Multichannel
2-D	Audio data that has been pre-processed with a Fourier transform: We can transform the audio waveform into a 2-D tensor with different rows corresponding to different frequencies and different columns corresponding to different points in time. Using convolution in the time makes the model equivariant to shifts in time. Using convolution across the frequency axis makes the model equivariant to frequency, so that the same melody played in a different octave produces the same representation but at a different height in the network's output	Color image data: One channel contains the red pixels, one the green pixels, and one the blue pixels. The convolution kernel moves over both the horizontal and the vertical axes of the image, conferring translation equivariance in both directions

Structured Outputs and Data Types IV

	Single channel	Multichannel
3-D	Volumetric data: A common source of this kind of data is medical imaging technology, such as CT scans.	Color video data: One axis corresponds to time, one to the height of the video frame, and one to the width of the video frame.

Advanced Topics I

1. Efficient Convolution Algorithms

- ▶ Using the outer product of the vector for the kernel (separable kernel).

2. Random or Unsupervised Features

- ▶ Non-learning feature extractor.
- ▶ Random initialization.
- ▶ Design by hand (edge operator, etc).
- ▶ Unsupervised type: Clustering with patches of image, and using centroid as the kernel.

Advanced Topics II

- ▶ Random: Used to choose the best architecture before learning.
- ▶ Greedy-approach: Step by step learning.
- ▶ Today, most convolutional networks are trained in a purely supervised fashion.

Advanced Topics III

4. The Neuroscientific Basis for Convolutional Networks

► V1: Cartoon view

1. V1 is arranged in a spatial map: CNN's sparse connection.
2. V1 contains many simple cells: Detector unit of CNN.
3. V1 also contains many complex cells: invariant pooling, cross-channel pooling, maxouts.

► Grandmother cells and infero temporal cortex (ITC).

1. When viewing an object, information flows from the retina, through the LGN, to V1, then onward to V2, then V4, then IT.
2. Layers of CNN are similar to this flow.

Advanced Topics IV

- ▶ Difference between CNN and brain.
 1. Fovea (brain): Focusing region having high resolution whereas others have low resolutions (equal resolutions and attention mechanism).
 2. Integrated with other sensors (brain): CNNs are purely visual.
 3. Much more than just recognize objects (brain): Recently, CNNs are used for the multi-modal data (but premature).
 4. Feedback (V1 of brain): CNNs are not.
 5. Intermediate regime between V1 and ITC (brain): More complicated than CNN, but not well-known.
- ▶ Neuroscience has told us relatively little about how to train convolutional networks.
- ▶ The use of back-propagation to train time-delay neural networks (TDNNs) for time series (successful).

Advanced Topics V

- ▶ **Reverse correlation:** In neuroscience, white noise images are revealed to find activated neurons (implying Gabor functions).
- ▶ **Gabor function:**

$$s(I) = \sum_{x \in \mathbb{X}} \sum_{y \in \mathbb{Y}} w(x, y) I(x, y)$$

- $w(x, y \mid \alpha, \beta_x, \beta_y, f, \phi, x_0, y_0, \tau) = \alpha \exp(-\beta_x x'^2 - \beta_y y'^2) \cos(fx' + \phi)$

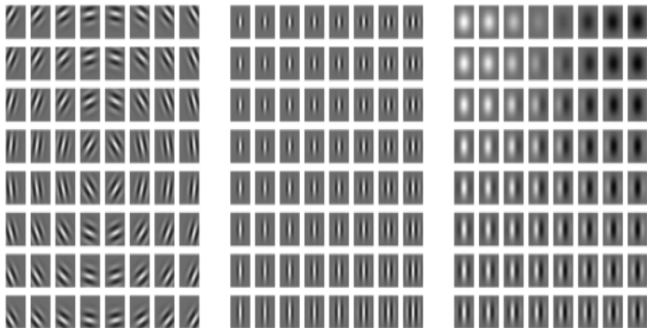
where

$$x' = (x - x_0) \cos(\tau) + (y - y_0) \sin(\tau)$$

$$y' = -(x - x_0) \sin(\tau) + (y - y_0) \cos(\tau)$$

- Gaussian and cosine is the main terms of Gabor function.

Advanced Topics VI

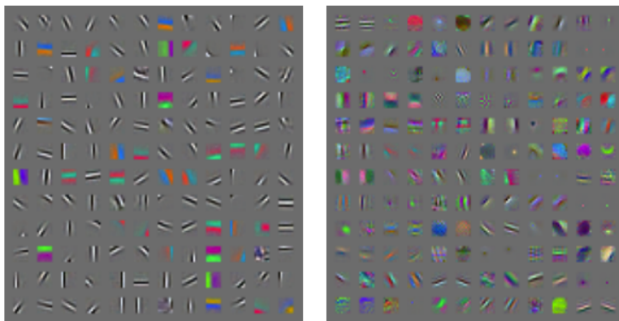


Various Gabor functions.

Advanced Topics VII

- ▶ The scaling factor α adjusts the total magnitude of the simple cell's response, while β_x and β_y control how quickly its receptive field falls off.
- ▶ The cosine factor $\cos(fx' + \phi)$ controls how the simple cell responds to changing brightness along the x axis. The parameter f controls the frequency of the cosine, and ϕ controls its phase offset.
- ▶ Assume that s_0 and s_1 are Gabor functions w.r.t. two axes. When s_1 has all the same parameters as s_0 except for ϕ , and ϕ is set such that s_1 is one quarter cycle out of phase with s_0 .
- ▶ The complex cell is invariant to small translations of the image in direction, or to negating the image (replacing black with white and vice versa).

Advanced Topics VIII



The feature detectors are reminiscent of the Gabor functions known to be present in the primary visual cortex. (Left) Weights learned by an unsupervised learning algorithm (spike and slab sparse coding) applied to small image patches. (Right) Convolution kernels learned by the first layer of a fully supervised convolutional maxout network. Neighboring pairs of filters drive the same maxout unit.